

Parameter Identification for DTMCs

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1 Introduction

Discrete-time Markov chains (DTMCs) are models for systems that evolve by transitioning between a finite set of states according to some probabilities. They find applications in fields like bioinformatics, financial analysis, weather forecasting and many more. In this project, we focus on the classical two-state DTMC with states A and B , characterized by transition probabilities $p = P(A \rightarrow B)$ and $q = P(B \rightarrow A)$, and an unknown initial distribution. Our goal is to apply different statistical methods to estimate these parameters from observed state-sequences.

2 Preliminaries and Specific Objectives

Consider the two-state Markov chain with transition matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

Suppose we observe a finite number N of finite traces (length n) our goal is to estimate the parameters p , q , and the initial distribution $\pi_A = \mathbb{P}(s_0 = A)$ using different parameter inference techniques:

- **Maximum-Likelihood Estimation (MLE):** Maximize the joint probability of the observed transitions to obtain $\hat{p}$, $\hat{q}$, and $\hat{\pi}_A$.
- **Bayesian Inference:** Place Beta priors on p , q , π_A , and derive the posterior from the observed transitions.

3 Methods

3.1 Maximum Likelihood Estimation

Since the N traces are independent, and each trace is generated by a time-homogeneous Markov chain, the likelihood of the parameters p , q and π_A given the observed data is:

$$\mathcal{L}(\text{data} \mid \pi_A, p, q) = \prod_{i=1}^N \Pr(s_0^{(i)}) \prod_{t=0}^{n-2} \Pr(s_t^{(i)} \rightarrow s_{t+1}^{(i)}).$$

Because

$$\Pr(s_0^{(i)}) = \pi_A^{\mathbf{1}\{s_0^{(i)}=A\}} (1 - \pi_A)^{\mathbf{1}\{s_0^{(i)}=B\}}$$

and

$$\Pr(s_t^{(i)} \rightarrow s_{t+1}^{(i)}) = \begin{cases} p, & \text{if } s_t^{(i)} = A, s_{t+1}^{(i)} = B, \\ 1-p, & \text{if } s_t^{(i)} = A, s_{t+1}^{(i)} = A, \\ q, & \text{if } s_t^{(i)} = B, s_{t+1}^{(i)} = A, \\ 1-q, & \text{if } s_t^{(i)} = B, s_{t+1}^{(i)} = B, \end{cases}$$

we can rewrite the full product as

$$\mathcal{L} = \prod_{i=1}^N [\pi_A^{\mathbf{1}\{s_0^{(i)}=A\}} (1 - \pi_A)^{\mathbf{1}\{s_0^{(i)}=B\}} \prod_{t=0}^{n-2} p^{\mathbf{1}\{s_t^{(i)}=A, s_{t+1}^{(i)}=B\}} (1-p)^{\mathbf{1}\{s_t^{(i)}=A, s_{t+1}^{(i)}=A\}} q^{\mathbf{1}\{s_t^{(i)}=B, s_{t+1}^{(i)}=A\}} (1-q)^{\mathbf{1}\{s_t^{(i)}=B, s_{t+1}^{(i)}=B\}}]$$

Now if we define the following quantities in this way:

$$\sum_{i=1}^N \mathbf{1}\{s_0^{(i)} = A\} = N_A^0, \quad \sum_{i=1}^N \mathbf{1}\{s_0^{(i)} = B\} = N_B^0,$$

and similarly

$$\begin{aligned} \sum_{i,t} \mathbf{1}\{s_t^{(i)} = A, s_{t+1}^{(i)} = A\} &= N_{AA}, & \sum_{i,t} \mathbf{1}\{s_t^{(i)} = A, s_{t+1}^{(i)} = B\} &= N_{AB} \\ \sum_{i,t} \mathbf{1}\{s_t^{(i)} = B, s_{t+1}^{(i)} = A\} &= N_{BA}, & \sum_{i,t} \mathbf{1}\{s_t^{(i)} = B, s_{t+1}^{(i)} = B\} &= N_{BB}, \end{aligned}$$

The likelihood becomes:

$$\mathcal{L}(\text{data} \mid \pi_A, p, q) = \pi_A^{N_A^0} (1 - \pi_A)^{N_B^0} p^{N_{AB}} (1 - p)^{N_{AA}} q^{N_{BA}} (1 - q)^{N_{BB}}.$$

and the corresponding log-likelihood is:

$$\log \mathcal{L} = N_A^0 \log \pi_A + N_B^0 \log(1 - \pi_A) + N_{AB} \log p + N_{AA} \log(1 - p) + N_{BA} \log q + N_{BB} \log(1 - q)$$

To obtain the parameter estimates we maximize the log-likelihood, so we take derivatives and solve:

$$\begin{aligned} \frac{\partial}{\partial \pi_A} \log \mathcal{L} &= \frac{N_A^0}{\pi_A} - \frac{N_B^0}{1 - \pi_A} = 0 \Rightarrow \hat{\pi}_A = \frac{N_A^0}{N} \\ \frac{\partial}{\partial p} \log \mathcal{L} &= \frac{N_{AB}}{p} - \frac{N_{AA}}{1 - p} = 0 \Rightarrow \hat{p} = \frac{N_{AB}}{N_{AA} + N_{AB}} \\ \frac{\partial}{\partial q} \log \mathcal{L} &= \frac{N_{BA}}{q} - \frac{N_{BB}}{1 - q} = 0 \Rightarrow \hat{q} = \frac{N_{BA}}{N_{BA} + N_{BB}} \end{aligned}$$

3.2 Bayesian Estimation

Another way to obtain estimates for the parameters of the model is to treat them probabilistically and assign independent prior distributions on each one. If the priors are Beta distributions, then the posteriors of the parameters are still Beta distributions given the form of the Likelihood.

$$\pi_A \sim \text{Beta}(\alpha_\pi, \beta_\pi), \quad p \sim \text{Beta}(\alpha_p, \beta_p), \quad q \sim \text{Beta}(\alpha_q, \beta_q).$$

Hence the posteriors are

$$\begin{aligned} \pi_A \mid \text{data} &\sim \text{Beta}(\alpha_\pi + N_A^0, \beta_\pi + N_B^0) \\ p \mid \text{data} &\sim \text{Beta}(\alpha_p + N_{AB}, \beta_p + N_{AA}) \\ q \mid \text{data} &\sim \text{Beta}(\alpha_q + N_{BA}, \beta_q + N_{BB}) \end{aligned}$$

From these distributions we can obtain both point estimates such as the posterior mean or median, and credibility intervals for the parameters.

4 Results

4.1 Maximum Likelihood Estimates

To assess how the number of observations affects estimator quality and to validate our methods under known parameters, we carried out a Monte Carlo experiment. We fixed the “true” values

$$p_{\text{true}} = 0.3, \quad q_{\text{true}} = 0.2, \quad \pi_A^{\text{true}} = 0.8,$$

and simulated $M = 500$ independent replications for each choice of $N \in \{50, 100, 300\}$ sample paths of length $n = 25$. In each replication:

1. We generated N traces by

- drawing the initial state s_1 as A with probability π_A^{true} ,
- evolving the chain for $t = 1, \dots, n-1$ using the transition matrix $P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$

2. We calculated the following quantities $N_{AA}, N_{AB}, N_{BA}, N_{BB}$ and N_A^0 .

3. We computed the MLEs

$$\hat{p} = \frac{N_{AB}}{N_{AA} + N_{AB}}, \quad \hat{q} = \frac{N_{BA}}{N_{BA} + N_{BB}}, \quad \hat{\pi}_A = \frac{N_A^0}{N}.$$

By standard asymptotic theory, each MLE is approximately normal for large total transitions $T = N(n-1)$:

$$\hat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{N_{AA}+N_{AB}}\right), \quad \hat{q} \approx \mathcal{N}\left(q, \frac{q(1-q)}{N_{BB}+N_{BA}}\right), \quad \hat{\pi}_A \approx \mathcal{N}\left(\pi_A, \frac{\pi_A(1-\pi_A)}{N}\right).$$

So we can compute the empirical 95% intervals as $\bar{\theta} \pm 1.96 SD(\theta)$ for $\theta \in \{\hat{p}, \hat{q}, \hat{\pi}_A\}$ and check how they change as N grows. The plots below show the empirical distributions of the estimators obtained through Monte Carlo simulation.

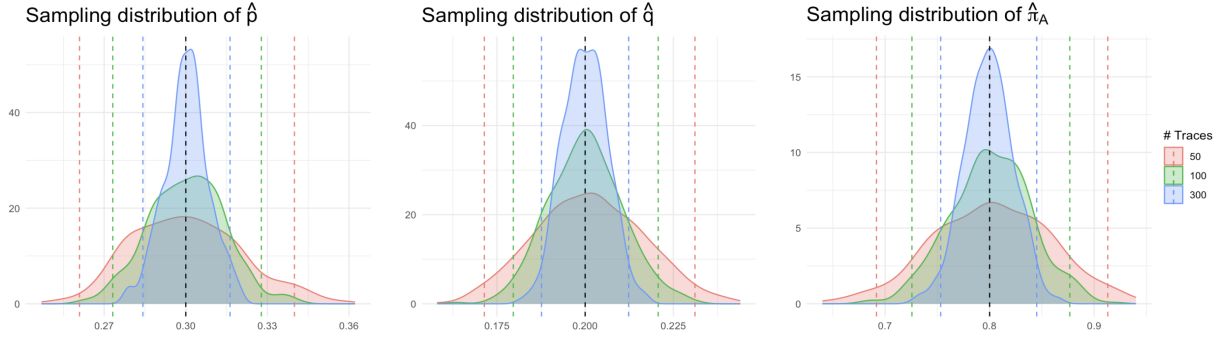


Figure 1: Density estimates (with 95% CIs) for p , q and π_A at different sample sizes. The dashed black line indicates the true value of the parameter.

The variance of the estimates of the parameters decreases linearly with the number of observations in each sample, which is shown in the picture by the fact that at higher sample sizes the empirical distributions are more concentrated around the true values of the parameters.

4.2 Bayesian Estimates

To assess the quality of the Bayesian estimators, we simulate N traces of length n . We obtain posterior distributions on the parameters of the model varying N between $\{500, 1000, 3000\}$ while keeping n fixed to 25, the results are shown in the figures below. All prior distributions are $Beta(2, 2)$, while the true values of the parameters are the same as in the MLE section.



Figure 2: Posteriors (with 95% CIs) for p , q and π_A at different sample sizes. The dashed black line indicates the true value of the parameter.

We can use the posterior distribution to obtain different estimators of the parameters, such as the posterior mean, the posterior median, and credible intervals, which provide a range within which the parameter values lie with a specified probability (0.95 in our experiments).

From Figure 2, it is evident that as N increases, the posterior densities concentrate more sharply around the true parameter values. So increasing the sample size not only centers the posterior mean closer to the ground truth but also reduces the posterior spread, yielding more reliable parameter estimates. The code relative to the simulations can be found [here](#).

5 General DTMC parameter estimation

Just like the form of the Likelihood function for a two-state DTMC is the product of binomial distributions, the general form of the Likelihood with transition probability matrix P and initial distribution P_0 is the product of multinomial distributions (one for each row of the transition matrix, plus one for the initial distribution).

$$\mathcal{L}(\text{data} \mid P_0, P) = \prod_{i=1}^N Pr(s_0^{(i)}) \prod_{t=0}^{n-2} Pr(s_t^{(i)} \rightarrow s_{t+1}^{(i)})$$

Equating the gradient of the function with respect to (P_0, P) to zero, we can obtain the Maximum Likelihood Estimators for the parameters of the model. This yields

$$\hat{P}_0(j) = \frac{C_j}{N} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{s_0^{(i)} = j\}$$

for the initial distribution, and

$$\hat{P}_{ij} = \frac{\sum_{l=1}^N \sum_{t=0}^{n-2} \mathbf{1}\{s_t^{(l)} = i, s_{t+1}^{(l)} = j\}}{\sum_{l=1}^N \sum_{t=0}^{n-2} \mathbf{1}\{s_t^{(l)} = i\}} = \frac{N_{ij}}{\sum_{j'=1}^k N_{ij'}}$$

for the transition matrix. This result holds when each $P_{i,j}$ is a free real number. More generally, when different parametrizations are chosen, the log-likelihood becomes a sum of logarithms of nonlinear functions which is generally nonconvex. As a result, there is no simple count-based closed-form solution, and one must use numerical optimization to find the maximum-likelihood estimate under the parametric form.